MIL-G-16748A
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SUPERSEDING
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MILITARY SPECIFICATION

GAGE, TANK LIQUID LEVEL (STATIC-HEAD)

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.—This specification covers tank liquid level gages of the static-head type intended for, but not restricted to, shipboard use.

1.2 Classification.—Tank liquid level gages (static-head) shall be of the following types, as specified (see 6.2):

Type A—Low-pressure tank liquid level gage.

Type B—High-pressure tank liquid level gage.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591 —Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).

NN-B-601 —Boxes, Wood-Cleated-Plywood, for Domestic Shipment.

NN-B-621 —Boxes, Wood, Nailed and Lock-Corner.

NN-B-631 —Boxes; Wood, Wirebound (for Domestic Shipment).

LLL-B-631 —Boxes; Fiber, Corrugated (for Domestic Shipment).

LLL-B-636 —Boxes, Fiber, Solid (for Domestic Shipment).

MILITARY

MIL-P-14 —Plastic-Materials, Molding, and Plastic Parts, Molded; Thermosetting.

JAN-P-105 —Packaging and Packing for Overseas Shipment — Boxes; Wood, Cleated Plywood.

JAN-P-106 —Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.

JAN-P-108 —Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board) Exterior and Interior.

MIL-P-116 —Preservation, Methods of.

JAN-P-133 —Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.

MIL-A-140 —Adhesive, Water-Resistant, Waterproof Barrier-Material.

MIL-M-191 —Mercury.

MIL-D-963 —Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).

MIL-C-972 —Compound, Rust-Preventive, Thin-Film (Polar Type).

MIL-L-10547 —Liners, Case, Waterproof.

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MIL–E–15090—Enamel, Equipment, Light Gray (Formula No. 111).

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Parts coming in contact with mercury shall be of glass tubing, plastic-material, molded, or iron, which shall be thoroughly pickled, followed by neutralizing, cleaning, and coating, with enamel baked on. Other metal parts of the gage shall be of such material, or so treated, as to resist corrosion when in contact with salt-laden air. Plastic-material, molded, if used, shall conform to type CFI of Specification MIL–P–14, except that electrical insulation tests will not be required. Piping interconnecting indicators, pumps, and tanks shall be of seamless copper tubing protected the full length by suitable brass or copper flexible tubing, or by being encased in a flexible flameproof cable made up as follows:

Asbestos filler shall be laid between the copper tubes to obtain a circular cross section and then covered with black felted asbestos material, thoroughly impregnated with black, moisture-resistant compound. The whole shall then be covered with black asbestos braid, finished smooth by use of a waxed covering. The final outside diameter and surface finish shall be such that a watertight joint may be obtained when the cable is installed by passing through Navy standard electric cable terminal tubes at bulkheads, the terminal tubes having an inside diameter of $^{15}/_{32}$, $^{21}/_{32}$, or $^{3}/_{4}$ inch. The material of piping and fittings within the tank shall resist chemical action on the fluid to be measured. When liquid used in the gage is other than mercury, as provided for in 3.6, the various parts of the gage in contact with the liquid shall be of materials resistant to chemical reaction with the liquid used.

3.1.2 Materials not specifically defined herein shall be of the best commercial quality.

3.1.3 Tank-top fittings and air bells shall be made of brass or bronze containing at least 85 percent copper.

3.2 Tank liquid level gages (static-head) shall operate by obtaining a balance with air between a column of liquid in a gage and the liquid in the tank.

3.3 All gages shall indicate correctly, regardless of any air pressure which may exist above the surface of the liquid, within the maximum pressure specified in 3.13.2 and 3.14.2 for the respective type of gage. The design shall permit installation of the piping interconnecting the tank and gage by the shortest possible route and, when the gage is not located at a level higher than the tank top, shall not require that this piping be led to a level more than 12 inches higher than the top of the tank proper (see 6.2). The design shall be such that inadvertent overflowing of the tank, or creation of an air pressure above the surface of the liquid in the tank, will not cause a loss of the indicating liquid in the gage. The system shall include an equalizing tube from top of tank to top of gage. Where the installation is such that inadvertent overflowing of the tank will not cause the loss of the indicating medium from the gage, or air pressure above atmospheric on the surface of the liquid in

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the tank cannot be created, an equalizing line between the gage and tank will not be required.

3.4 The tank liquid level gage shall consist of the following:

(a) An indicating gage.
(b) Air tubing with necessary connections.
(c) Flexible metallic tubing.
(d) Bulkhead and deck fittings.
(e) Tank-top fitting.
(f) Am air bell, or equivalent thereof, within the tank.
(g) For type A gages, a manually operated air pump, built integral with or mounted with the gage.
(h) For type B gages, a suitable operating valve to permit blowing out the air bell tubing and equalizing tubing without subjecting the gage to the imposed pressure.
(i) For type B gages, a manually operated air pump built integral with or mounted with the gage, as specified (see 6.2) or a valve arranged to admit air from a compressed-air system, as specified (see 6.2).

3.5 Unless otherwise specified in the contract or order, the interconnecting piping between indicator pump, and tank shall be so arranged as not to require manipulation of valves or cocks, other than required by 3.4 and 3.14.4, between tank and gage, either for adjusting of the zero level of the liquid in the air bell or for taking readings. A valve may be used at discharge side of pump.

3.6 Unless otherwise specified in contract or order, the gage shall consist of a vertical fixed liquid column, fitted with two metal scales, the whole inclosed in a metal frame, suitable for wall mounting, with front glass and metal retainer. The case and frame shall be painted with light gray enamel conforming to class 2 of Specification MIL–E–15090, except that stainless steel parts shall not be painted. Unless otherwise specified (see 6.2), the indicating liquid shall be mercury conforming to grade II of Specification MIL–M–191.

3.7 Means shall be provided for filling the mercury reservoir without disassembling the gage.

3.8 The glass tubing shall be "white-black" tubing or plain tubing backed with white "pyralin" or similar material. The minimum bore of the glass tubing shall be not less than one-eighth inch in diameter. The gage shall be so designed and assembled as to permit ready replacement of broken gage glasses. The mercury reservoir shall have a uniform inside cross-sectional area not less than 20 times the area of the bore of the glass tube.

3.9 The design of the air bell or pipe, where the air bell consists of a pipe, shall be such that when the air in the bell and the pipes leading to both the gage and the pump or air valve is compressed from atmospheric pressure to the maximum pressure for which the system is desined, as specified in 3.13.2 and 3.14.2, by an imposed liquid level in the tank or an imposed pressure on top of the liquid, the maximum height to which the liquid within the bell will rise shall not reach to within 1 inch of the terminus of the tube connecting to the bell or pipe. The above assumes no leakage of air from the piping system.

3.10 When an air pump is provided it shall be of adequate capacity, and means shall be provided to prevent the escape of air from the system through the pump.

3.11 All fittings for connecting the piping shall be furnished. The tank-top fitting for the equalizing air line shall be designed so as to be provided with means satisfactory to the bureau or agency concerned to prevent the entry of the tank liquid into the equalizing air line. Holes in the tank-top fitting shall be straight to permit cleaning with a drill from outside the tank. The air chamber

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shall not extend below the inside of the tank-top plating, except that the bell may project into the tank where the tank top abuts on an armored deck. Bulkhead and deck fittings shall incorporate stuffing boxes for air tubing and means for securing ends of flexible metallic protective tubing or stuffing boxes for the protective tubing if other than flexible metallic.

3.12 The seamless tubing and the flexible metallic protective tubing interconnecting the gage, pump, and valve with the tank shall be furnished in continuous lengths, as specified (see 6.2). Joints in the seamless tubing shall be made with approved cone or compression fittings. Joints in the flexible metal tubing shall be made with approved fittings. All tubes leading from a tank to a gage, pump, and valve shall be led in a single protective tube. The size of the interconnecting tubing outside of the tank shall not exceed one-fourth inch outside diameter. For tubing over one-tenth inch inside diameter, the connection from the pump and/or valve shall be made to the gage-pressure line at the gage. For tubing one-tenth inch or less inside diameter, the connection from the pump and/or valve may be made to the gage-pressure line at the tank top. The size of the tubing within the tank connecting to the air bell shall be not less than three-eighths inch inside diameter.

3.13 Type A, low-pressure tank liquid level gage.

3.13.1 The manually-operated air pump shall be used for expelling liquid from the air bell.

3.13.2 The gage shall be designed to withstand an internal pressure of 20 pounds per square inch gage and temperatures up to 150° F.

3.1.3.3 *Scales.* — The scales shall be installed on each side of the mercury column. All markings shall be indelibly marked on the scales.

3.13.3.1 *Right-hand scales.*

3.13.3.1.1 Right-hand scales shall be calibrated in gallons on the basis of the liquid which is to be measured at 60° F., and using the average specific gravity of that liquid, and graduated to read in hundreds of gallons. They shall be marked as follows:

> Graduations with numerals shall be placed at intervals of approximately 1 inch, at the nearest even-hundred-gallons location. The numerals shall indicate only hundreds of gallons. The intervals between consecutive graduations shall be subdivided by graduations, without numerals, to indicate each hundred gallons. The gallon mark corresponding to tank capacity of "zero" of the air bell shall be distinctly indicated and labeled as follows: "Zero of air bell ____ inches above the lowest point of tank."

3.13.3.1.2 The right-hand scales shall be installed on the gages with the above-prescribed mark opposite the zero inch mark of the left-hand scales.

3.13.3.2 *Left-hand scales.*

3.13.3.2.1 Left-hand scales shall be calibrated to represent pressure in inches of mercury (or other liquid, if specified (see 3.6)) and graduated to read in inches and tenths, and shall be heavily scribed to show levels corresponding to 95 percent of full capacity of tank with liquids of 0.90, 0.95, and 1.00 specific gravity, respectively.

3.13.3.2.2 The length of the scales shall be such as is necessary to accommodate them to the tank for which constructed. All scales shall be inscribed with the ship's number of the tank for which the gage is constructed. The scales shall be finished in dull white with markings filled in black. Data for calibration purposes will be furnished by the Government.

3.13.4 The requirements of 3.13.1 to 3.13.3 inclusive, are for gages for fuel-oil tanks on surface vessels. Type A gages for tanks containing other liquids, or for specal types of

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service, shall comply with the same requirements, except that the calibration and graduations of the scales specified in 3.13.3 shall be as specified (see 6.2).

3.14 Type B, high-pressure tank liquid level gage.

3.14.1 A manually-operated air pump, built integral with or mounted with the gage, or a valve arranged to admit air from a compressed-air system, shall be provided for expelling liquid from the air bell.

3.14.2 The gage shall be designed for, and shall function satisfactorily under tank blow pressures of from 0 to 150 pounds per square inch, under compartment pressures (on the gages) of from 0 to 20 pounds per square inch, and under temperatures of 20° to 150° F. The gages shall be of the mercury-column type or approved dial type, as specified (see 6.2).

3.14.3 In the case of mercury-column gages, the scales shall be installed on each side of the mercury column. All markings shall be indelibly marked on the scales. The length of the scales shall be such as is necessary to accommodate them to the tank for which the gage is constructed. The scales shall be finished in dull white with markings filled in black. The left-hand scale shall be calibrated to represent pressure in inches of mercury or water, and graduated to read in inches and tenths. Unless otherwise specified in the contract or order, the right-hand scale shall be left blank for calibration after delivery. The scale shall also bear the inscription "Zero point of air bell above lowest point of tank ____ inches," and "Tank No. ____." These data will be filled in when the right-hand scale is calibrated. The scales of dial-type gages shall be calibrated as specified (see 6.2).

3.14.4 *Operating valve.*

3.14.4.1 The operating valve specified in 3.4 shall be so designed that the plug can be set in four different positions.

3.14.4.1.1 In one position the tank air bell shall be connected through a port to the compressed air (hand pump or ships service air) connection of the valve, simultaneously the top and bottom of the gage shall be connected through another port to a vent from the valve.

3.14.4.1.2 By an angular clockwise movement of not more than 30°, a second position of the plug shall connect through a port, the equalizer line from the tank top with the compressed air connection, simultaneously all other connections shall be closed off.

3.14.4.1.3 By a second progressive clockwise movement of 30° (third position), the plug shall close off all gage, tank, and vent connections and shall be designated as the "OFF" position.

3.14.4.1.4 By a third and final clockwise movement of 30° (fourth position), the plug shall by the arrangement of ports connect the bottom (pressure side) of the gage with the air bell and the top of the gage through the equalizer line to the tank top, simultaneously the value vent and compressed air connection shall be flanked.

3.14.4.2 Each position of the valve plug shall be clearly labeled to indicate the valve function at the respective positions and stops shall be provided at the extremes of travel. Connections and ports for a valve are shown on figure 1.

3.14.4.3 The bosses of the connections on the valve body shall be stamped as shown on figure 1 to facilitate installation.

3.15 Repair parts.—Repair parts, as applicable to the respective manufacturer's product, shall be furnished as follows:

For air valve

Compression spring—25 percent of total order but not less than 1.

For gage

Gage glass—100 percent of order.
Spring for gage mounting posts—25 percent of order but not less than 1.

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Indicating fluid (mercury or other type)—200 percent of order.

For air pump

Valve spring—25 percent of order but not less than 1.

Plunger leather—100 percent of order.

3.16 **Drawings.**—Drawings showing overall dimensions, material list and estimated weight shall accompany bids. Drawings in accordance with Specification MIL–D–963 shall be furnished as specified in the contract or order.

3.17 **Workmanship.** — The workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Inspection procedures.**—For naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 **Inspection tests at place of manufacture.**

4.2.1 All equipment shall be inspected by the Government inspector to determine compliance with this specification, as to material, design, workmanship, and the dimensions specified on the applicable drawings.

4.2.2 Each gage, together with valves, pump, and other appliances pertaining thereto shall be tested hydrostatically to one and one-half times the maximum pressure specified for the type of gage.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.**

5.1.1 *For domestic shipment and storage.*—Gages (and pump if attached) shall be wrapped in creped cellulose wadding or similar suitable nonabrasive material and shall be packaged individually in set-up boxes or corrugated or solid-fiberboard boxes conforming to Specification JAN–P–133 or JAN–P–108, respectively. Air tubing and flexible metallic tubing shall be coiled and secured with suitable twine, tape, or wire ties. Fittings shall be packaged in suitable kraft envelopes or bags. Air bells shall be wrapped in not less than 60 pound minimum basis weight kraft paper, secured with suitable tape. Pumps (if separate unit) and valves for type B gages shall be packaged individually in set up boxes conforming to Specification JAN–P–133.

5.1.2 *For oversea shipment.*—Component parts of gages shall be packaged as specified in 5.1.1, except that surfaces subject to corrosion shall first be thoroughly cleaned, prepared, preserved, and wrapped in accordance with method I or I–A as applicable, of Specification MIL–P–116. The corrosion-preventive compound used shall conform to grade I, II, or III, as applicable of Specification MIL–C–972.

5.2 **Packing.**

5.2.1 *For domestic shipment and storage.*—Unless otherwise specified in the contract or order, complete gages with component parts, packaged as specified in 5.1.1, shall be packed in wood-cleated-fiberboard, wood-cleated-plywood, nailed wood, wirebound wood, corrugated fiber, or solid fiber boxes conforming to Specification NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, or LLL–B–636, respectively. The gross weight of wood or wood-cleated boxes shall not exceed approximately 150 pounds. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification.

5.2.2 *For oversea shipment.*—When specified (see 6.2), complete gages with component parts, packaged as specified in 5.1.2 and with suitable interior packing to prevent shifting or damage from shock, shall be individually packed in cleated plywood boxes or nailed wood boxes conforming to Specification JAN–P–105 or JAN–P–106, respectively. Shipping containers shall be fitted with a sealed case liner conforming to type I,

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grade B, class 2 of Specification MIL–L–10547. Seams and closures shall be sealed with adhesive conforming to Specification MIL–A–140. Shipping containers shall be strapped in conformance with the appendix of the applicable box specification. The gross weight of the boxes shall not exceed approximately 100 pounds.

5.2.3 *For domestic or oversea shipment.*—Where practicable, shipping containers shall be of uniform size, and shall be designed to fit the contents in a compact manner.

5.3 Marking.—In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall be in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use.

6.1.1 Type A, low pressure gages are intended for use where the maximum pressure imposed on the gage does not exceed 20 pounds per square inch (gage).

6.1.2 Type B, high pressure gages are intended for use with tank blow pressures from 0 to 150 pounds per square inch and compartment pressures (on the indicators) from 0 to 20 pounds per square inch.

6.2 Ordering data.—Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type required (see 1.2).

(c) The total depth of tank, the distance of the indicating gage from the tank by the nearest permissible route (see 3.3).

(d) Where a manually operated air pump is required, whether it is built integral with or mounted with the indicating gage (see 3.4).

(e) In the case of type B gages, whether a manually operated air pump or a valve arranged to admit air from a compressed air system is required (see 3.4).

(f) The indicating liquid required if other than mercury (see 3.6).

(g) The lengths in which the tubing shall be supplied (see 3.12).

(h) In the case of type A gages, the calibration and graduations of scales where indicating liquid is not mercury or the systems are for special types of service and the liquid to be measured with the average specific gravity at 60° F. (see 3.13.4).

(i) Whether mercury-column or dial type gages are required for type B gages (see 3.14.2); and if dial type, the calibration of the scales (see 3.14.3).

(j) Whether domestic or oversea shipment is required (see 5.1 and 5.2).

6.3 Superseding documents.—This specification supersedes MIL–G–16748(SHIPS). The latter superseded Navy Specifications 45–I–2.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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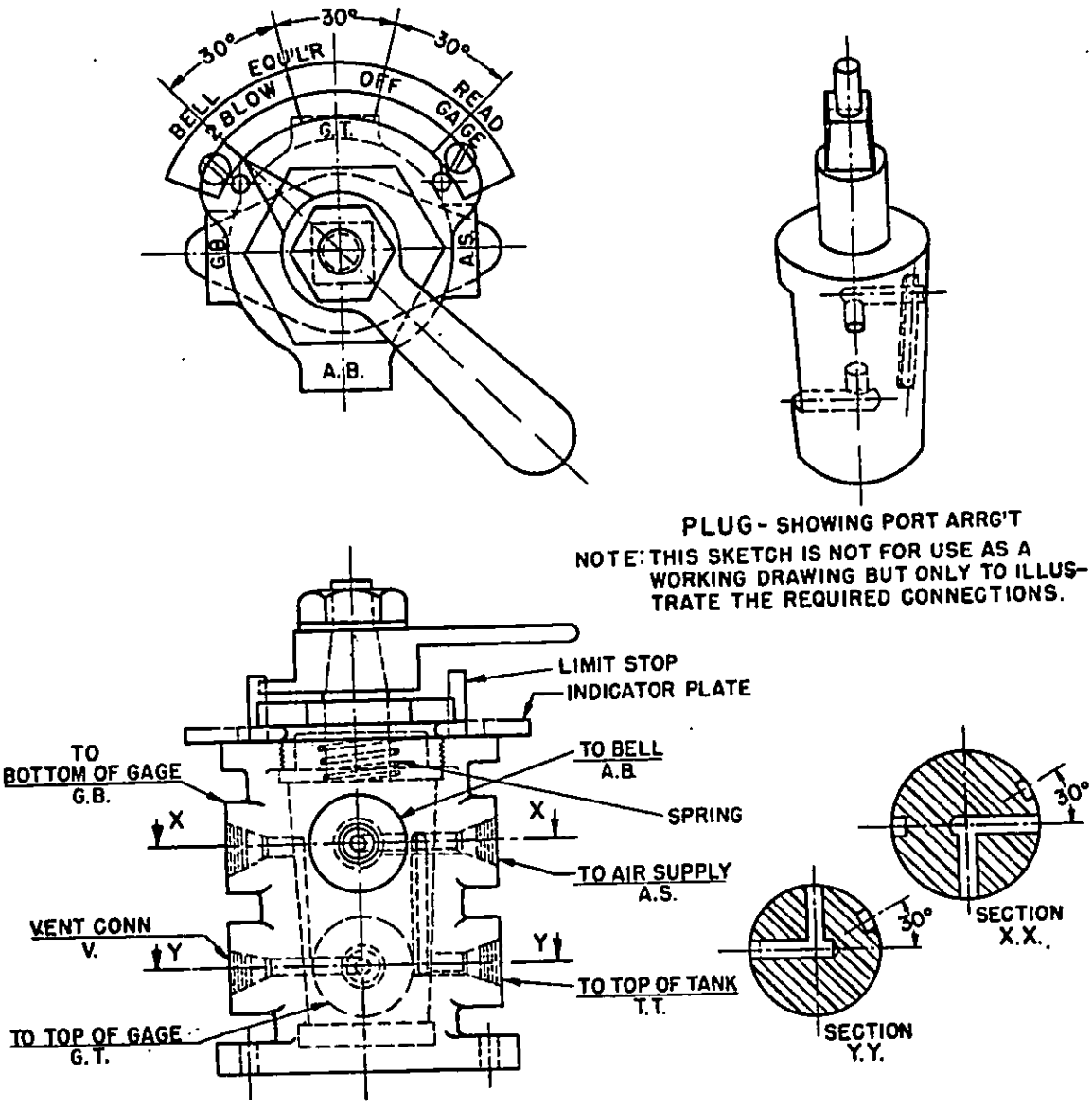


FIGURE 1.—*Plug valve.*

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